# Marcus Feldman

Research Scientist | Machine Learning Infrastructure

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## PROFESSIONAL SUMMARY

ML systems researcher with 6 years of experience designing and evaluating compiler optimization frameworks and runtime systems for deep learning workloads. PhD in Computer Science focused on automatic differentiation infrastructure. Published research on tensor scheduling and memory allocation strategies with 340+ combined citations. Skilled at building benchmarking pipelines and evaluating novel architectures on standardized datasets.

## EXPERIENCE

**Research Scientist, ML Infrastructure Group**

*Cascade Research Institute, Portland, OR* | Jan 2024 – Present

- Developed TensorBind, an automatic scheduling framework for heterogeneous accelerators; published results on MLPerf and SPEC benchmarks showing 2.3× improvement over baselines
- Designed evaluation methodology for comparing 12 different tensor layout strategies across MNIST, ImageNet, and WikiText datasets
- Authored 3 peer-reviewed papers on memory hierarchy optimization; primary publication received 89 citations within 18 months of publication

**PhD Candidate, Computer Science**

*University of Washington, Seattle, WA* | Sept 2019 – Dec 2023

- Dissertation: "Adaptive Gradient Computation Graphs for Mixed-Precision Training"
- Published 5 papers in top venues (NeurIPS, ICML, ICLR) covering automatic differentiation systems and compilation strategies
- Implemented reverse-mode AD compiler in LLVM; benchmarked against PyTorch and JAX on synthetic operator graphs
- Accumulated 251 citations across all publications

**Graduate Research Assistant**

*University of Washington* | Sept 2019 – Dec 2023

- Built reproducible evaluation harness for gradient accumulation patterns on simulated hardware
- Contributed foundational work on cost models for tensor operations

## EDUCATION

**PhD in Computer Science**, University of Washington, 2023

**BS in Computer Science**, Reed College, Portland, OR, 2019